

Probability Distributions and Loss Functions in Machine Learning

(Bishop Chapter 2 — Handout)

A. Likelihood Models (Data Distributions)

These distributions model *observed data*.

Data type	Distribution	pmf / pdf	Mean, Variance
Binary	Bernoulli(μ)	$\mu^x(1-\mu)^{1-x}$	$\mathbb{E}[x] = \mu, \text{Var}(x) = \mu(1-\mu)$
Counts	Binomial(N, μ)	$\binom{N}{x}\mu^x(1-\mu)^{N-x}$	$N\mu, N\mu(1-\mu)$
Categorical	Categorical(π)	$\prod_k \pi_k^{x_k}$	$\mathbb{E}[x] = \pi$
Real-valued	Gaussian(μ, σ^2)	$\mathcal{N}(x \mu, \sigma^2)$	μ, σ^2
Positive real	Exponential(λ)	$\lambda e^{-\lambda x}$	$1/\lambda, 1/\lambda^2$

B. Prior Distributions (Parameter Uncertainty)

These distributions model uncertainty about parameters.

Likelihood	Parameter	Conjugate prior
Bernoulli / Binomial	μ	Beta(a, b)
Categorical / Multinomial	π	Dirichlet(α)
Poisson	λ	Gamma(a, b)
Gaussian mean	μ	Gaussian
Gaussian precision	λ	Gamma

Beta:

$$\text{Beta}(\mu | a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1-\mu)^{b-1}.$$

Dirichlet:

$$\text{Dir}(\pi | \alpha) = \frac{1}{B(\alpha)} \prod_{k=1}^K \pi_k^{\alpha_k - 1}, \quad \sum_k \pi_k = 1.$$
$$\mathbb{E}[\pi_k] = \frac{\alpha_k}{\alpha_0}, \quad \alpha_0 = \sum_k \alpha_k.$$

C. Likelihood $\rightarrow$ Loss Functions

Negative log-likelihood induces loss functions.

Gaussian Likelihood

$$y = w^T x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

$$-\log p(y | x, w) = \frac{1}{2\sigma^2} (y - w^T x)^2 \Rightarrow \text{Mean Squared Error.}$$

Convex quadratic loss.

Bernoulli Likelihood

$$p(y \mid x, w) = \sigma(w^T x)^y (1 - \sigma(w^T x))^{1-y}.$$

$$-\log p(y \mid x, w) = -y \log \hat{y} - (1 - y) \log(1 - \hat{y}).$$

Binary cross-entropy loss.

Convex in prediction; convex in w for logistic regression.

Categorical Likelihood

Softmax model:

$$p(y = k \mid x) = \frac{e^{w_k^T x}}{\sum_j e^{w_j^T x}}.$$

$$\mathcal{L} = -\log p(y \mid x) \Rightarrow \text{Multiclass cross-entropy.}$$

D. Likelihood $\rightarrow$ Sufficient Statistics $\rightarrow$ MLE

Model	Sufficient statistic	MLE
Bernoulli	$\sum x_n$	$\hat{\mu} = \frac{1}{N} \sum x_n$
Gaussian (μ)	$\sum x_n$	sample mean
Gaussian (μ, Σ)	$\sum x_n, \sum x_n x_n^T$	mean, covariance
Poisson	$\sum x_n$	sample mean

Key principle:

MLE depends on data only through sufficient statistics.

E. Bayesian Inference and MAP

Bayes' rule:

$$p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)p(\theta).$$

MAP estimator:

$$\theta_{\text{MAP}} = \arg \max_{\theta} [\log p(\mathcal{D} \mid \theta) + \log p(\theta)].$$

MAP $\equiv$ regularized optimization.

F. Summary Structure

$$\text{Likelihood} \rightarrow \begin{cases} \text{NLL} \rightarrow \text{loss function} \\ \text{sufficient statistics} \rightarrow \text{MLE} \end{cases}$$

$$\text{Bayesian: Prior} \rightarrow \text{Posterior} \rightarrow \text{MAP} / \text{Prediction}$$